Week 4

MICROECONOMIC THEORY
ECON 323 502/503

Spring 2026

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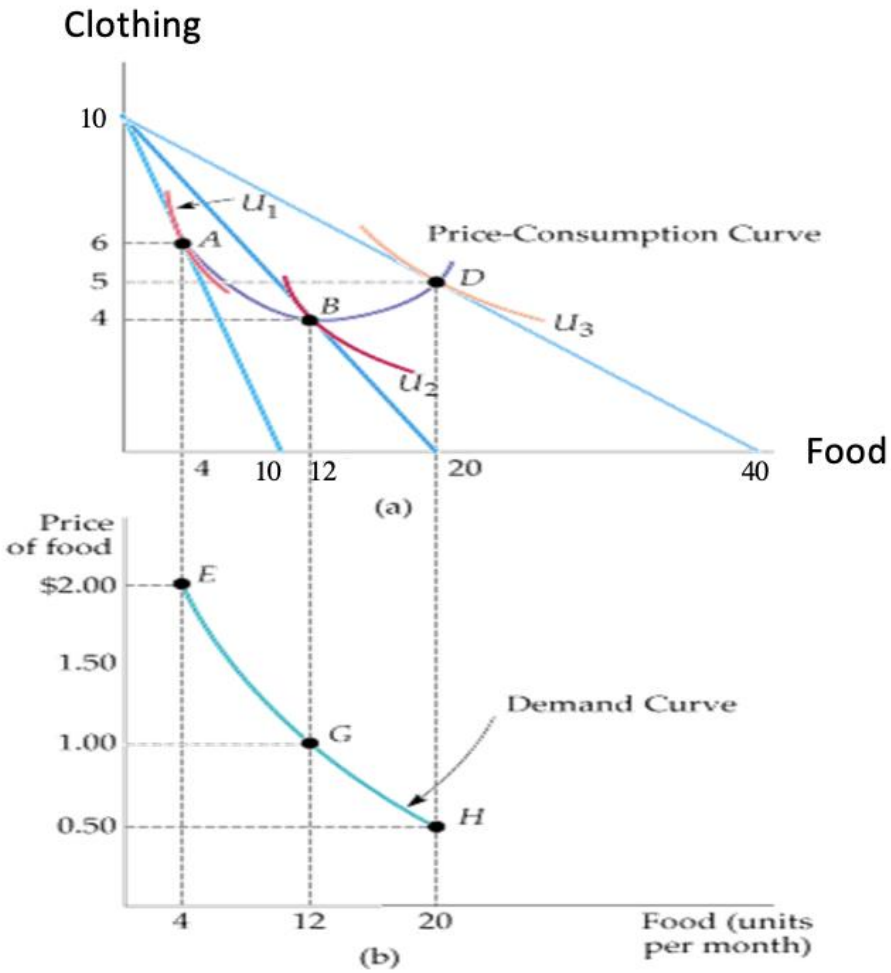
Review

Analyze how P_X and I affect the utility maximizing (x, y) – the graphical approach

To describe the **relationship between P_X and x**:

Step 1. We can plot **price-consumption curve** first.

Step 2. Construct **demand curve**.



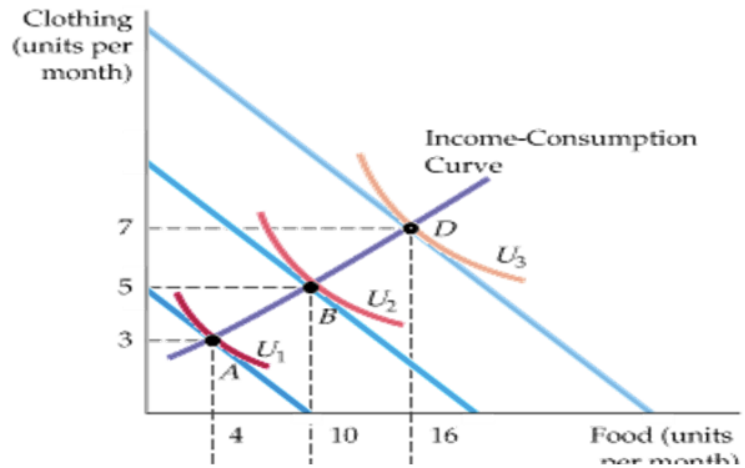
Review

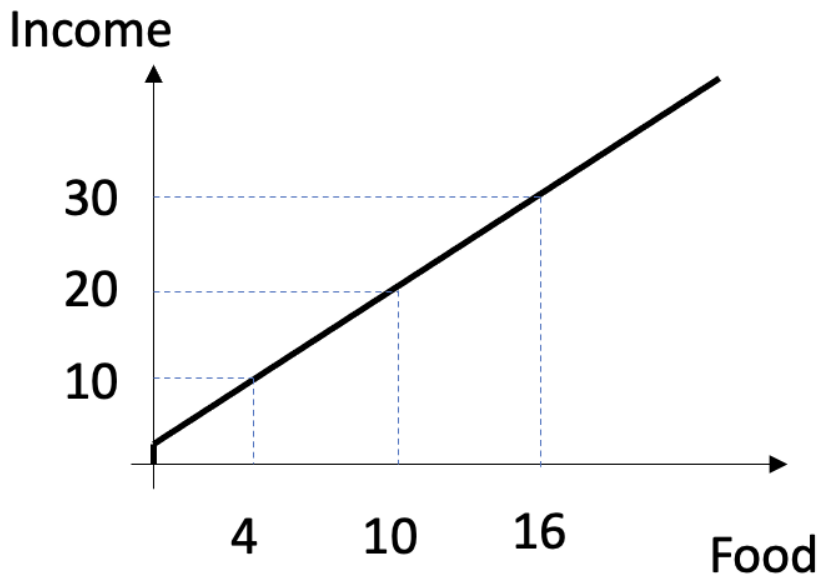
To describe the **relationship between I and x**:

Step 1. Plot **income-consumption curve** first.

Step 2. Construct **Engel curve**.

- Normal Good (an increase in I leads to an increase in x)
- Inferior Good (an increase in I leads to a decrease in x)





Derive the demand curve – the analytical approach

Step 1: Fix P_Y and I as constant numbers.
Step 2: Solve the utility maximization problem solution x as a function of P_X. Rename the solution x as $Q_X(P_X)$.

Derive the Engel curve – the analytical approach

Step 1: Fix P_X and P_Y as constant numbers.
Step 2: Solve the utility maximization problem solution x as a function of I. Rename the solution x as $Q_X(I)$.

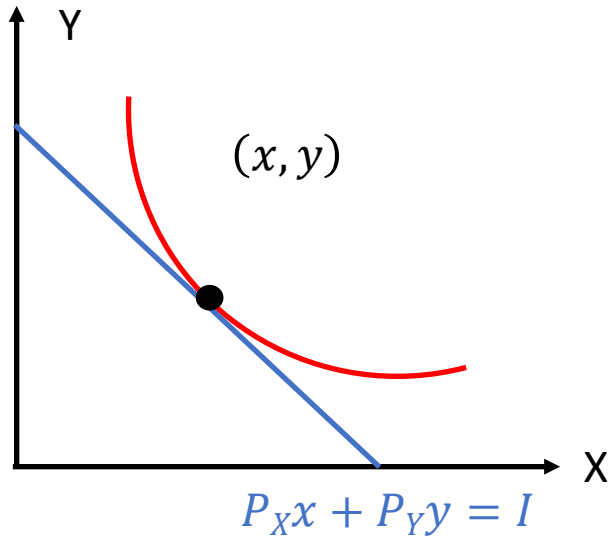
Individual Demand Curve and Engel Curve: Analytical Solution

Example 1: Consider a problem with diminishing MRS and smooth preference. Suppose $MRS = \frac{y}{x}$ everywhere in this problem.

a) Derive the demand curve of X when $P_Y = 10$ and $I = 100$.

b) Derive the Engel curve of X when $P_X = 20$ and $P_Y = 10$. Is good X normal or inferior?

Solution:



General utility maximization conditions for the default case: The utility maximizing bundle (x, y) must satisfy

$$MRS = \frac{P_X}{P_Y} \text{ (1) and } P_X x + P_Y y = I \text{ (2).}$$

Individual Demand Curve: Analytical Solution

Derive the demand curve of X when $P_Y = 10$ and $I = 100$. Recall $MRS = \frac{y}{x}$ in this problem.

From $MRS = \frac{P_X}{P_Y}$, we further have $\frac{y}{x} = \frac{P_X}{10}$ (1)

From $P_X x + P_Y y = I$, we further have $P_X x + 10y = 100$ (2).

Isolate y from (1), we have $y = \frac{P_X x}{10}$ (3).

Plug (3) into (2), we have $P_X x + 10\frac{P_X x}{10} = P_X x + P_X x = 2P_X x = 100$.

Solve for x, we have $x = \frac{100}{2P_X} = \frac{50}{P_X}$.

Hence, the demand curve of Good X is: $Q_X(P_X) = \frac{50}{P_X}$.

Engel Curve: Analytical Solution

Derive the Engel curve of X when P_X=20 and P_Y=*10*. Is good X normal or inferior? Recall $MRS = \frac{y}{x}$ in this problem.

From $MRS = \frac{P_X}{P_Y}$, we further have $\frac{y}{x} = \frac{20}{10} = 2$ (1)

From $P_X x + P_Y y = I$, we further have $20x + 10y = I$ (2).

Isolate y from (1), we have $y = 2x$ (3).

Plug (3) into (2), we have $20x + 10(2x) = 40x = I$.

Solve for x, we have $x = \frac{I}{40}$.

Hence, the Engel curve of Good X is: $Q_X(I) = \frac{I}{40}$.

Good X is normal because the demand is increasing in I.

Individual Demand: Analytical Solution

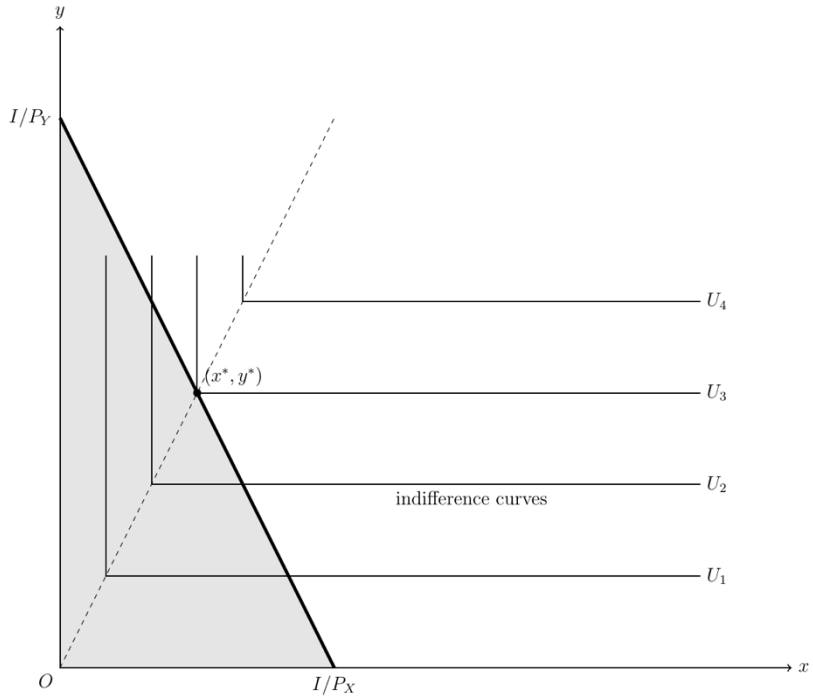
Example 2: Suppose good X and good Y are perfect complements and the kinks of indifference curve fall on $y = 2x$ line.

a) Derive the demand curve of X when $P_Y = 10$ *and* $I = 100$.

b) Derive the Engel curve of X when $P_X = 20$ and $P_Y = 10$. Is good X normal or inferior?

Individual Demand: Analytical Solution

Solution:



The utility maximizing market basket (x, y) satisfies

$P_X x + P_Y y = I$ and $y = 2x$.

Individual Demand: Analytical Solution

Derive the demand curve of X when P_Y *=10 and I=100.*

From $P_X x + P_Y y = I$ we have $P_X x + 10y = 100$ (1)

Recall that $y = 2x$ (2).

Plug (2) into (1), we have $P_X x + 10\,(2x) = P_X x + 20x = (P_X + 20)x = 100.$

Solve for x, we have $x = \frac{100}{P_X + 20}$ (3)

Hence, the demand curve of Good X is $Q_X(P_X) = \frac{100}{P_X + 20}$.

Individual Demand: Analytical Solution

Derive the Engel curve of X when P_X=20 and P_Y=*10*. Is good X normal or inferior?

From $P_X x + P_Y y = I$ we have $20x + 10y = I$ (1)

Recall that $y = 2x$ (2).

Plug (2) into (1), we have $20x + 10\,(2x) = 40x = I$.

Isolate x, we have $x = \frac{I}{40}$.

Hence, the Engel curve is given by $Q_X(I) = \frac{I}{40}$.

X is a normal good because when income increases, the consumption of x increases.

So far, we have graphically and analytically studied how quantity demanded of X is affected by P_X and I. Namely, we have learned how to derive the demand curve and the Engel curve.

There is another element that affects the quantity demanded of a good --- the price of the other good.

4. Individual Demand: Substitutes and Complements

Substitutes: Two goods are substitutes if an increase (resp. decrease) in the price of one good leads to an increase (resp. decrease) in the quantity demanded of the other good.

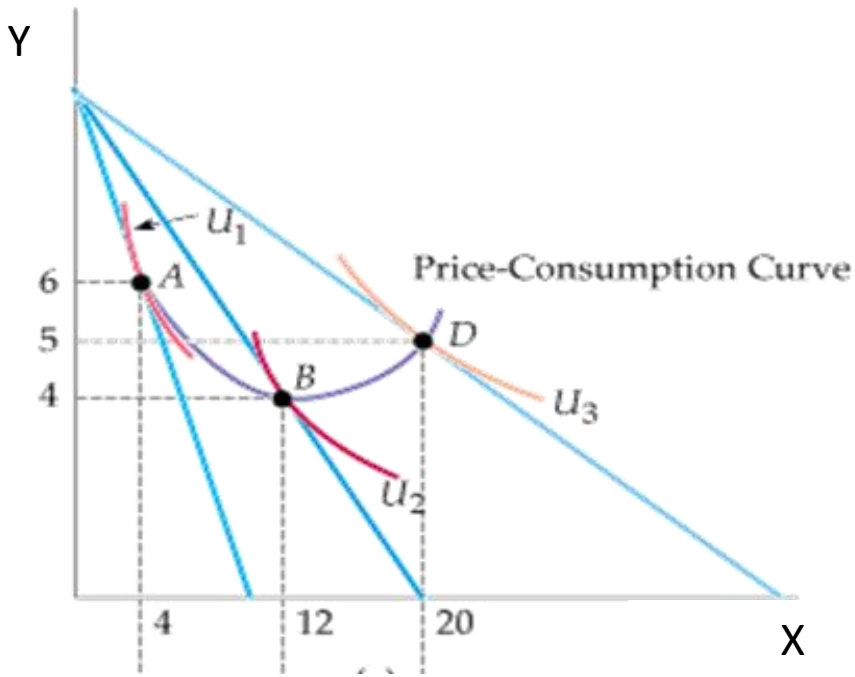
e.g., beef and pork, iMac and PC, gym membership and outdoor running

Complements: Two goods are complements if an increase (resp. decrease) in the price of one good leads to a decrease (resp. increase) in the quantity demanded of the other good.

e.g., movies and popcorns, ground beef and bun, iPhone and iPhone cases

Independent: The two goods are independent if an increase (resp. decrease) in the price of one good does not affect the quantity demanded of the other good.

4. Individual Demand: Substitutes and Complements



There are examples of other flavors, but in this example, when P_X increases (budget line becomes steeper):

- From D to B, quantity demanded of Y decreases. In this case, X and Y are complements.
- From B to A, quantity demanded of Y increases. In this case, X and Y are substitutes.

4. Individual Demand: Substitutes and Complements

How can two goods be substitutes sometimes and complements at other times?

- Think about the relationship between and iPad and MacBook.
- When price of iPad is low, iPad and MacBook may be complements – they work very well together.
- When price of iPad is high, iPad and MacBook may become substitutes – if I spent too much money on one, then I may be less interested in buying the other.